

miRNA-seq Analysis Pipeline for IBD

Dataset PRJNA471862 · Ulcerative Colitis · End-to-End Workflow

DATA ACQUISITION



00 Dataset Discovery & Metadata Curation

EDirect · SRA Toolkit [01_geo_ibd_curator.py, 02_bioproject_curator.py]

candidates_master.csv · runinfo.csv · dataset metadata

01 Raw Data Download

prefetch · fasterq-dump · pigz [03_download_runs.sh]

Raw reads · SRR*.fastq.gz (160 samples)

02 Quality Control

FastQC · MultiQC · miRTrace [04_qc_fastqc.sh, 06_qc_mirtrace.sh]

FastQC / MultiQC report · miRTrace QC report

03 miRNA Quantification

miRge3.0 · miRBase hsa [09_auto_subfolders_mirge3.sh, 10_single_folder_mirge3.sh]

miRNA.Counts.csv · miRNA.RPM.csv (per sample)

04 Count Merge & Filtering

pandas · mean RPM ≥ 100 filter [11_mrlna_cut_off.py, 12_deseq2_metadata.py]

Master_Filtered_Counts_DESeq2.csv · sample_metadata.csv

05 Differential Expression Analysis

DESeq2 · apeglim · VST · PCA / UMAP [13_run_dea.R]

DEA results · Default & Strict models

PCA · UMAP · Heatmap · Volcano

06 Target Prediction

multiMiR · miRTarBase · TarBase · miRecords [14_multimir_targets.R]

targets_validated.csv · targets_predicted.csv

07a Pathway Enrichment

clusterProfiler · GO · KEGG [15_pathway_enrich.R]

GO_Enrichment.csv · KEGG_Enrichment.csv · dotplots

07b Network, Clinical & Database Validation

igraph · ROC · OpenTargets · HMDD / miRNet [16_network.R, 17_tr_response_roc.R, 18_hmdd_prep.R, 19_ibd_validation.R]

network HTML · ROC/AUC curves · HMDD lists · IBD overlap

08 Automated HTML Report & Interactive Dashboard

Plotly.js · vis.js · matplotlib [20_generate_report.py, 16_interactive_network.R]

PubMed / NCBI Literature DB

PubMed Search
search_engine_papers/

Paper Retrieval
& Deduplication

Local Search Engine
& Corpus Indexing

Filtered corpus
(papers.csv / .jsonl)

context

SEQUENCING PROCESSING

STATISTICAL ANALYSIS

BIOLOGICAL INTERPRETATION

REPORTING

Legend

- Data Acquisition (00-01)
- Quality Control (02)
- Quantification & Processing (03-04)
- Statistical Analysis (05)
- Biological Interpretation (06-07)
- Literature Mining (parallel)
- Intermediate data / files
- Data flow
- Supplementary link